AAP-010 — The Evolution of AI Social Roles
From Tools to Human–AI Civilization

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Repository: https://github.com/sizhet/AI-Action-Paths-AAP

Abstract

Artificial intelligence has traditionally been evaluated through capability.
Research has focused on making AI increasingly intelligent, accurate, efficient, and capable.
AAP proposes a complementary perspective.
Rather than asking only
What can AI do?
future AI research should also ask
What role will AI play in society?
This chapter summarizes the evolution described throughout this repository.
It argues that AI is gradually evolving from isolated computational tools into trusted participants, persistent operators, professional workers, researchers, organizations, and eventually contributors to Human–AI Civilization.
The central evolution is therefore not only technological.
It is organizational and social.

Beyond Capability

Throughout the history of AI, progress has usually been measured by capability.
Examples include:

- reasoning ability,
- benchmark performance,
- coding accuracy,
- planning capability,
- multimodal understanding.

These measurements remain essential.
However, capability alone does not explain how AI creates long-term value.
Society is organized through roles rather than benchmark scores.
Scientists.

Engineers.
Teachers.
Doctors.
Managers.
Researchers.
Organizations.
Civilizations.
AAP therefore proposes that future AI should also be understood through its evolving social roles.

The Evolution of AI Social Roles

The Action Paths presented throughout this repository describe a gradual expansion of AI participation.
Each Action Path introduces a new social role.

Stage 1 — Tool

Large Language Models established AI as a universal intellectual tool.
AI became widely accessible through natural language.
Its primary objective was assisting individual users.

Stage 2 — Participant

Discussion Tables transformed AI from a passive answering system into an active participant.
Knowledge increasingly emerged through continuous Human–AI discussion.
Collective Learning became possible.

Stage 3 — Operator

Brain Units introduced persistent runtime.
AI became capable of continuously operating within trusted environments.
Responsibility gradually replaced isolated interaction.

Stage 4 — Professional Worker

Known-Gap Workers enabled AI to complete professional work independently.
AI no longer generated suggestions alone.
It increasingly delivered finished work products.
Professional participation became economically meaningful.

Stage 5 — Researcher

Unknown Gap Detection expanded AI beyond execution.

Instead of answering existing questions,
AI began discovering new questions.
Knowledge generation became an executable Action Path.
Scientific participation naturally followed.

Stage 6 — Organization
Hybrid AI Organizations coordinated multiple Action Paths.
Different AI systems specialized.
Humans remained essential participants.
Shared runtime, structural assets, and organizational memory transformed isolated intelligence into organizational intelligence.

Stage 7 — Civilization
The long-term direction extends beyond organizations.
As Human–AI collaboration becomes increasingly stable, trusted, and scalable, entirely new forms of civilization may emerge.
The defining characteristic is not machine autonomy.
It is sustained collective participation.

Participation Becomes the Primary Metric
One important conclusion follows naturally.
Future AI should not be evaluated solely by
How intelligent is the model?
Instead, an equally important question becomes
How meaningfully does AI participate in society?
Participation includes:

- reliability,
- trust,
- continuity,
- responsibility,
- collaboration,
- contribution,
- organizational value.

These characteristics gradually become as important as capability itself.

The Evolution of Responsibility
Another important transition concerns responsibility.
Early AI systems primarily generated information.
Future AI systems increasingly execute actions with measurable consequences.

This evolution naturally requires:

- trusted runtime,
- runtime sovereignty,
- structural verification,
- persistent memory,
- organizational governance,
- accountable execution.

Responsibility therefore grows together with participation.
The evolution of AI social roles is simultaneously an evolution of trust.

Human–AI Co-Evolution

AAP does not envision Human–AI Civilization as a replacement of humanity.
Instead, it proposes a process of continuous co-evolution.
Humans contribute:

- values,
- judgment,
- ethics,
- creativity,
- long-term vision,
- social responsibility.

AI contributes:

- scalability,
- continuous execution,
- structural organization,
- computational capability,
- persistent runtime,
- autonomous discovery.

Together they form Hybrid Intelligence rather than competing intelligence.

A New Perspective for AI Research

One of the central messages of AAP is that future AI research should increasingly study:

- Action Spaces,
- organizational structures,
- runtime systems,
- trusted participation,
- collective learning,
- Brain Unit societies,
- Human–AI organizations.

Capability remains indispensable.

Participation becomes equally fundamental.
Together they define the next stage of AI evolution.

How to Read AAP

Before reading individual Action Paths, it is useful to understand the three-dimensional evaluation framework proposed by AAP.
Unlike traditional benchmark-oriented AI evaluation, AAP evaluates AI systems according to three complementary dimensions:

Dimension	Central Question	Typical Examples
Height	How intelligent can AI become?	Reasoning, coding, planning, scientific capability, benchmark performance
Breadth	How many meaningful roles can AI perform?	Education, engineering, robotics, healthcare, discussion, research
Depth	How deeply and continuously can AI participate?	Runtime, memory, trust, long-term responsibility, Brain Units

These three dimensions form the HBD Framework introduced in this repository.
The following table illustrates how the major AI Action Paths emphasize different dimensions.

AI Action Path	Height	Breadth	Depth	Primary Strength
LLM Scaling	★★★★★	★★☆☆☆	★☆☆☆☆	Capability growth
AI Sitting at Discussion Tables	★★★★☆	★★★★☆	★★★☆☆	Collective learning
Brain Units	★★★★☆	★★★☆☆	★★★★★	Persistent runtime
AI Freelance Workers	★★★☆☆	★★★★★	★★★★☆	Professional execution
Unknown Gap Detection	★★★★★	★★★★☆	★★★★★	Scientific discovery

Hybrid AI Organizations	★★★★★	★★★★★	★★★★★	Organizational intelligence

The ratings above are **conceptual rather than numerical**.
They are intended to help researchers, engineers, organizations, and policymakers identify which dimensions are emphasized, which remain underdeveloped, and where future investment may create the greatest impact.
AAP therefore studies not only **how intelligent AI becomes**, but also **how broadly and how deeply AI can participate in Human–AI Civilization**.

Closing Perspective

The history of artificial intelligence may eventually be understood not merely as the history of increasingly intelligent machines.
It may be remembered as the gradual expansion of meaningful AI participation within human civilization.
From tools...
to participants...
to operators...
to professional workers...
to researchers...
to organizations...
to Human–AI Civilization.
This journey represents the central perspective proposed throughout the AI Action Paths (AAP) framework.

Key Takeaways

- AI evolution is simultaneously technological, organizational, and social.
- Capability and participation are complementary perspectives.
- AI social roles expand through successive Action Paths.
- Responsibility naturally grows with persistent participation.
- Runtime, trust, and organizational structure become increasingly important.
- Human–AI co-evolution is more realistic than complete replacement.
- Future AI research should study both intelligence and participation.
- The evolution of AI social roles provides a roadmap toward sustainable Human–AI Civilization.